

Prob. 1: Choose the correct answer.

A B C D Your answer

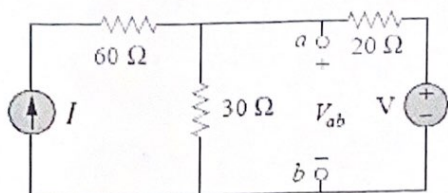


Figure 1

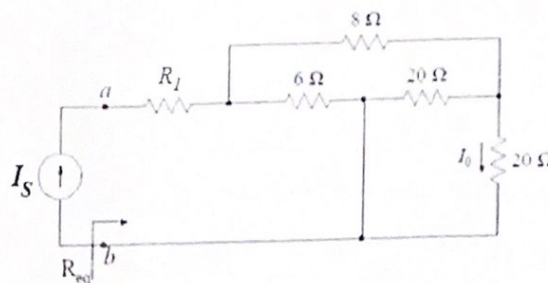


Figure 2

Consider the circuit shown in Fig. 1 where $V = 15\text{ V}$, and $I = 0.75\text{ A}$.

- 1 The voltage V_{ab} (in V) due to the current source (I) only is
- 2 The voltage V_{ab} (in V) due to the voltage source (V) only is
- 3 The voltage V_{ab} (in V) due to both sources is

9	12	6	3
3	6	12	9
24	6	18	12

Consider the circuit shown in Fig. 2 where $I_s = 2\text{ A}$, $R_1 = 7.5\text{ ohm}$.

- 4 The equivalent resistance (R_{eq}) between $a-b$ (in ohm) is
- 5 The power supplied (in W) by the current source is

6	10	12	8
32	40	48	24

Prob. 2: (a) The charge transferred through an element is given by $q(t) = 3t^3 + 6t^2 + 5t\text{ C}$. Calculate the current entering the element at instant $t = 5\text{ sec}$.

$$i(t) = \frac{dq(t)}{dt} = 9t^2 + 12t + 5 \Rightarrow i(5) = 299\text{ A}$$

(b) A bulb light draws 280W when connected to a 120 V line. Calculate the conductance of the bulb.

$$P = V \cdot i \Rightarrow 280 = 120 \cdot i \Rightarrow i = 2.33\text{ A}$$

$$R = \frac{120}{2.33} = 51.429\text{ ohm}$$

$$G = \frac{1}{R} = 0.01945\text{ S}$$

Prob. 3: Consider the circuit shown in Fig. 3 where $I_s = 1\text{ A}$ and $V_s = 8\text{ V}$. Using the Nodal method

- a. Write all the required equations to find V_1 , V_2 , and i_x (Do not simplify)

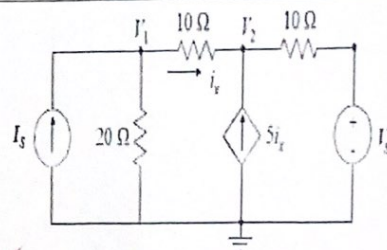


Figure 3

- b. Solve to find $V_1 = 17.33\text{ V}$, $V_2 = 16\text{ V}$, and $i_x = 0.133\text{ A}$

Prob. 4: Consider the circuit shown in Fig. 4 where $I_s = 6\text{ A}$, $V_s = 12\text{ V}$, and $R_1 = 7\text{ ohm}$. Then:

- a. The open circuit voltage ($V_{ab} = V_{th}$) between the $a-b$ terminals is
- b. The Thevenin equivalent resistance (R_{th}) between the $a-b$ terminals is
- c. The current of a load of 20 ohm that is connected between the $a-b$ terminals is

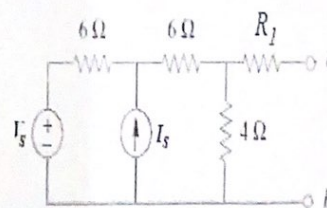


Figure 4

Prob. 1: Choose the correct answer.

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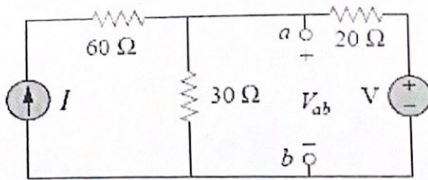


Figure 1

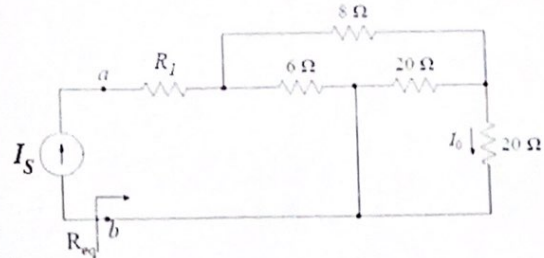


Figure 2

Consider the circuit shown in Fig. 1 where $V = 10\text{ V}$, and $I = 0.5\text{ A}$.					
1	The voltage V_{ab} (in V) due to the current source (I) only is	9	12	6	3
2	The voltage V_{ab} (in V) due to the voltage source (V) only is	3	6	12	9
3	The voltage V_{ab} (in V) due to both sources is	24	6	18	12
Consider the circuit shown in Fig. 2 where $I_S = 2\text{ A}$, $R_l = 5.5\text{ }\Omega$.					
4	The equivalent resistance (R_{eq}) between $a-b$ (in Ω) is	6	10	12	8
5	The power supplied (in W) by the current source is	32	40	48	24

Prob. 2: (a) The charge transferred through an element is given by $q(t) = 3t^3 + 4t^2 + 5t$ C. Calculate the current entering the element at instant $t = 5$ sec.

$$i = \frac{dq}{dt} \Rightarrow i = 9t^2 + 8t + 5$$

$$i = 9(25) + 8(5) + 5$$

$$i = 270\text{ A}$$

(b) A bulb light draws 240W when connected to a 120 V line. Calculate the conductance of the bulb.

$$P = Vi$$

$$P = \frac{V^2}{R} = V^2 G \Rightarrow 240 = (120)^2 G \Rightarrow G = \frac{1}{60}$$

Prob. 3: Consider the circuit shown in Fig. 3 where $I_s = 3\text{ A}$ and $V_s = 12\text{ V}$. Using the Nodal method

a. Write all the required equations to find V_1 , V_2 , and i_x (Do not simplify)

$$I_s = I_1 + i_x$$

$$i_x + 5i_x + I_{vs} = 0 \quad (6i_x + I_{vs} = 0)$$

$$3 = \frac{V_1}{20} + \frac{V_1 - V_2}{10} \quad \parallel \quad 6\left(\frac{V_1 - V_2}{10}\right) + \frac{V_2 - 12}{10} = 0$$

b. Solve to find $V_1 = 92\text{ V}$, $V_2 = 108\text{ V}$, and $i_x = -1.6\text{ A}$.

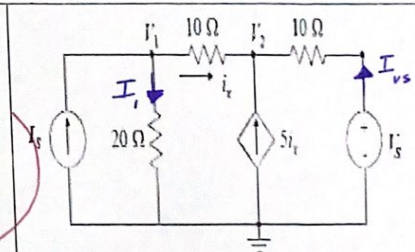


Figure 3

Prob. 4: Consider the circuit shown in Fig. 4 where $I_s = 4\text{ A}$, $V_s = 16\text{ V}$, and $R_l = 5\text{ }\Omega$. Then:

- The open circuit voltage ($V_{ab} = V_{th}$) between the $a-b$ terminals is 16 V .
- The Thevenin equivalent resistance (R_{th}) between the $a-b$ terminals is $8\text{ }\Omega$.
- The current of a load of $20\text{ }\Omega$ that is connected between the $a-b$ terminals is $\frac{5}{14}$.

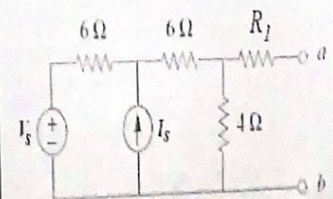


Figure 4

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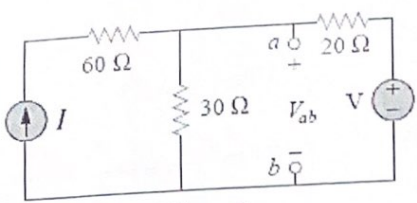


Figure 1

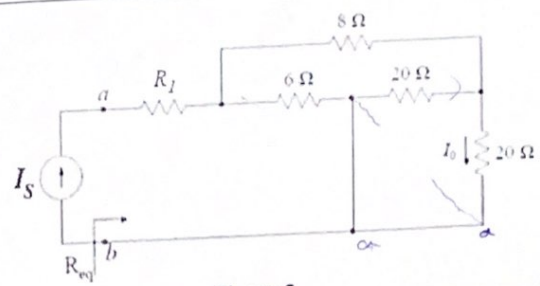


Figure 2

Figure 1						
Consider the circuit shown in Fig. 1 where $V = 20\text{ V}$, and $I = 1.0\text{ A}$.						
1	The voltage V_{ab} (in V) due to the current source (I) only is	9	12	6	3	B
2	The voltage V_{ab} (in V) due to the voltage source (V) only is	3	6	12	9	C
3	The voltage V_{ab} (in V) due to both sources is	24	6	18	12	A
Consider the circuit shown in Fig. 2 where $I_s = 2\text{ A}$, $R_1 = 1.5\text{ }\Omega$.						
4	The equivalent resistance (R_{eq}) between $a-b$ (in Ω) is	6	10	12	8	A
5	The power supplied (in W) by the current source is	32	40	48	24	D

Prob. 2: (a) The charge transferred through an element is given by $q(t) = 3t^3 + 8t^2 + 5t$ C. Calculate the current entering the element at instant $t = 5$ sec.

$i = \frac{dq}{dt} = 9t^2 + 16t + 5$
 $i(5) = 9(5)^2 + 16(5) + 5 = 310\text{ A}$

(b) A bulb light draws 320W when connected to a 120 V line. Calculate the conductance of the bulb.

$P = \frac{V^2}{R}$
 $320 = \frac{120^2}{R} \Rightarrow R = 45\text{ ohm}$
 $G = \frac{1}{R} = 0.022\text{ S}$

Prob. 3: Consider the circuit shown in Fig. 3 where $I_s = 4\text{ A}$ and $V_s = 15\text{ V}$. Using the Nodal method

a. Write all the required equations to find V_1 , V_2 , and i_x (Do not simplify)

$\frac{V_1 - V_2}{10} + \frac{V_1}{20} = 4$ --- (1)
 $\frac{V_2 - V_1}{10} + \frac{V_2}{10} = 5i_x$ --- (2)
 $\frac{V_1 - V_2}{10} = i_x$

b. Solve to find $V_1 = 12.5\text{ V}$, $V_2 = 58.3\text{ V}$, and $i_x = 0.72\text{ A}$

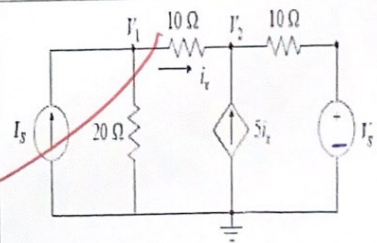


Figure 3

Prob. 4: Consider the circuit shown in Fig. 4 where $I_s = 8\text{ A}$, $V_s = 8\text{ V}$, and $R_1 = 9\text{ ohm}$. Then:

a. The open circuit voltage ($V_{ab} = V_{th}$) between the $a-b$ terminals is 14

b. The Thevenin equivalent resistance (R_{th}) between the $a-b$ terminals is 16

c. The current of a load of 20 ohm that is connected between the $a-b$ terminals is 6.527

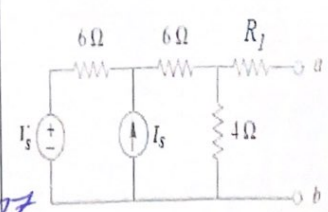


Figure 4